

Vadayath Jayakrishna Menon

Co-FOUNDER · ARTIPHISHELL INC

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Education

Arizona State University

Doctor of Philosophy in Computer Science

Arizona, USA

Aug 2019 - May 2026

- GPA: 4.0
- Research in Cyber Security

Amrita University

Bachelor of Technology in Computer Science and Engineering

Kerala, India

July 2014 - May 2018

- GPA: 9.27/10.0
- Graduated with Distinction

Work Experience

Artiphishell Inc

Co-Founder

California, USA

May 2026 - Current

- Co-founded Artiphishell Inc, a cybersecurity startup focused on automatic verification of security issues using AI.

Research Experience

SEFCOM Lab at Arizona State University

Research Assistant

Arizona, USA

Aug 2019 - May 2026

- Working with Dr. Yan Shoshitaishvili and Dr. Ruoyu "Fish" Wang
- Research topics: Binary Exploitation, Fuzzing, Symbolic Execution, Static Binary Analysis, Reverse Engineering

ForAllSecure

Intern

Pennsylvania, USA

May - Aug (2022) & May - Aug (2023)

- Worked with Thanassis Avgerinos and Maxwell Koo
- Worked on implementing a snapshot fuzzer for virtual machines

SEFCOM Lab at Arizona State University

Research Scholar

Arizona, USA

Oct 2018 - May 2019

- Worked under the supervision of Dr. Yan Shoshitaishvili and Dr. Ruoyu "Fish" Wang
- Analyzing binary application software for identifying vulnerabilities

ISI at University of Southern California

Research Scholar

California, USA

Feb 2018 - Oct 2018

- Worked under the supervision of Dr. Christophe Hauser
- Identifying potential leak of pointers through static binary analysis and symbolic execution

Research Publications

Fuzz to the Future: Uncovering Occluded Future Vulnerabilities via Robust Fuzzing

Salt Lake City, UT

31st ACM Conference on Computer and Communications Security (CCS 24)

2024

Arvind S Raj, Wil Gibbs, Fangzhou Dong, **Jayakrishna Menon Vadayath**, Michael Tompkins, Steven Wirsz, Yibo Liu, Zhenghao Hu, Chang Zhu, Gokulkrishna Praveen Menon, Brendan Dolan-Gavitt, Adam Doupé, Ruoyu Wang, Yan Shoshitaishvili and Tiffany Bao,

Operation Mango: Scalable Discovery of Taint-Style Vulnerabilities in Binary Firmware Services

Philadelphia, PA

33rd USENIX Security Symposium (USENIX Security 24)

2024

Wil Gibbs, Arvind S Raj, **Jayakrishna Menon Vadayath**, Hui Jun Tay, Justin Miller, Akshay Ajayan, Zion Leonahenahe Basque, Audrey Dutcher, Fangzhou Dong, Xavier Maso, Giovanni Vigna, Christopher Kruegel, Adam Doupé, Yan Shoshitaishvili, and Ruoyu Wang

"Watching over the shoulder of a professional": Why hackers make mistakes and how they fix them

San Francisco, CA

45th IEEE Symposium on Security and Privacy (S&P 24)

2024

Irina Ford, Ananta Soneji, Faris Bugra Kokulu, **Jayakrishna Vadayath**, Zion Leonahenahe Basque, Gaurav Vipat, Adam Doupé, Ruoyu Wang, Gail-Joon Ahn, Tiffany Bao, and Yan Shoshitaishvili,

Greenhouse: Single-Service Rehosting of Linux-Based Firmware Binaries in User-Space Emulation

Anaheim, CA

32nd USENIX Security Symposium (USENIX Security 23)

2023

Hui Jun Tay, Kyle Zeng, **Jayakrishna Menon Vadayath**, Arvind S Raj, Audrey Dutcher, Tejesh Reddy, Wil Gibbs, Zion Leonahenahe Basque, Fangzhou Dong, Zack Smith, Adam Doupé, Tiffany Bao, Yan Shoshitaishvili, and Ruoyu Wang

Arbiter: Bridging the Static and Dynamic Divide in Vulnerability Discovery on Binary Programs

Massachusetts, USA

31st USENIX Security Symposium (USENIX Security 22)

2022

Jayakrishna Vadayath, Moritz Eckert, Kyle Zeng, Nicolaas Weideman, Gokulkrishna Praveen Menon, Yanick Fratantonio, Davide Balzarotti, Adam Doupé, Tiffany Bao, Ruoyu Wang, Christophe Hauser and Yan Shoshitaishvili

Sleak: Automating Address Space Layout Derandomization

Puerto Rico, USA

The 35th Annual Computer Security Applications Conference

2019

Christophe Hauser, **Jayakrishna Menon**, Yan Shoshitaishvili, Ruoyu Wang, Giovanni Vigna and Christopher Kruegel

Professional Activities

IEEE Transactions on Dependable and Secure Computing

Reviewer

2026

Network and Distributed Systems Security Symposium

California, USA

Artifact Evaluation Committee Member

Oct 2026

ACM Conference on Computer and Communications Security

Utah, USA

Artifact Evaluation Committee Member

Oct 2024

18th USENIX WOOT Conference on Offensive Technologies (WOOT '24)

Pennsylvania, USA

Artifact Evaluation Committee Member

Aug 2024

Leadership Experience

Shellphish CTF Team

Co-Captain

2023 - 2026

- Led one of the world's top competitive CTF teams
- Coordinated team strategy and participation in international CTF competitions

Teaching Experience

CSE545: Software Security at Arizona State University

Arizona, USA

Teaching Assistant

Aug 2021- Dec 2021

- Assisted Dr. Ruoyu "Fish" Wang
- Primary duties were helping students with assignments and coursework

CSE365: Information Assurance at Arizona State University

Arizona, USA

Teaching Assistant

Aug 2020- Dec 2020

- Assisted Dr. Ruoyu "Fish" Wang
- Primary duties were helping students with assignments and coursework

Projects

Arbiter

A scalable and precise hybrid vulnerability analysis framework

- A hybrid vulnerability analysis framework that can be used to vulnerabilities that satisfy a given vulnerability description in binaries in a scalable and precise manner.
- Source code available on GitHub

PuzzIL

An intermediate language fuzzer for PHP

- An intermediate language fuzzer for the PHP JIT engine based on top of PhplL.
- Found 2 bugs in the JIT engine of PHP 8.0
- Source code available on GitHub

Vulnerability Research

2023	Wyze Camera 0-day , Pwn2Own	Vancouver
2022	Synology NAS 0-day , Pwn2Own	Vancouver
2018	CVE-2018-18311 , Heap overflow in Perl interpreter, CVSS 3.0 score: 9.8	

Extracurricular Activities

2025	Ranked 5th in AlxCC Finals with team Shellphish , DARPA AI Cyber Challenge	Las Vegas, USA
2024	AlxCC Finalist with team Shellphish , DARPA AI Cyber Challenge	Las Vegas, USA
2024	Small Business Track Competition Winners with team Shellphish , DARPA AI Cyber Challenge	
2023	11th rank globally with team Shellphish , DEFCON CTF Finals	Las Vegas, USA
2022	13th rank globally with team Shellphish , DEFCON CTF Finals	Las Vegas, USA
2021	10th rank globally with team Shellphish , HITB Pro CTF	Abu Dhabi, UAE
2020	7th rank globally with team Shellphish , DEFCON CTF Finals	Las Vegas, USA
2019	11th rank globally with team Pwndevils , Dragon CTF	Warsaw, Poland
2019	First runners up with team Pwndevils , CSAW Embedded Security Challenge	New York, USA
2016	First runners up with team bi0s , CSAW CTF 2016 (India)	Uttar Pradesh, India